

FRANCK JUNIOR ABOYA MESSOU

Full-Stack Engineer | Flutter Mobile · React Web · Node.js / Flask Backend | AI-Integrated Systems

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SUMMARY

Full-stack engineer shipping **cross-platform mobile apps (Flutter)**, **React web frontends**, and **Node.js / Flask backends** with GraphQL and REST. Background as a doctoral AI researcher gives sharp instincts for the data-engineering and ML-integration layers: sensor-data pipelines, model serving on dedicated GPU back-ends, and AI-augmented features inside production apps. Recent client work at D'isum Inc. (Tokyo) covered architecture, prototyping, and team integration across the full stack.

EXPERIENCE

Full-Stack Developer & AI (Part-time)

D'isum Inc.

 Oct 2025 – Jun 2026  Tokyo, Japan

- **Sensor Data Pipeline Architect & Analyst:** built ingestion-to-analytics pipelines from raw IoT sensor streams.
- **Sensor Data Specialist:** app development, prototyping, and integration across full scope of development, architecture, and team collaboration on core software integration.
- **Precision Sensor Data Engineer, Analytics, ML Prototypes & System Fusion:** integrated ML / predictive elements into client systems; prototyping and merging new components with existing infrastructure.
- **Sensor apps (iOS & Android):** built with Java and Swift.

Engineering Lead – EKKOU Platform

Independent / open-source

 2024 – Present  Tokyo, Japan

- Full-stack AI platform: **Flutter** mobile clients, **Flask / FastAPI** REST APIs, model serving on dedicated GPU back-ends.
- Owned end-to-end: data layer, API design, mobile UI, deployment, and CI / CD.

Doctoral Research Engineer

YuLab, Hosei University

 Oct 2025 – Present  Tokyo, Japan

- Engineering side of doctoral work: **provision and operate a dedicated multi-GPU server** (CUDA, NVIDIA drivers, DDP training, inference services).
- Research output: 2 Q1 journal papers (*Internet of Things, Expert Systems with Applications*) and 5 IEEE flagship conference papers.

EDUCATION

PhD, Applied Informatics

Hosei University

 2023 – Apr 2027 (early)  Tokyo, Japan

Advisor: Prof. Keping Yu (Stanford / Elsevier World's Top 2% Scientists 2025).

MSc, Applied Informatics

Hosei University

 2023 – Sep 2025  Tokyo, Japan

TECHNICAL SKILLS

Languages

Python Java Dart JavaScript TypeScript
SQL Bash

Mobile & Frontend

Flutter / Dart React HTML5 / CSS3
REST consumption GraphQL clients

Backend & APIs

Flask FastAPI Node.js / Express
GraphQL REST JWT / OAuth WebSockets

Databases

MongoDB PostgreSQL MySQL SQLite
Schema design & indexing

DevOps & Cloud

Docker docker-compose GitHub Actions
Make GCP Linux server admin

AI integration

PyTorch Hugging Face OpenAI / Claude APIs
vLLM Multi-GPU serving

Intern, GenAI for Business · TICAD9 Booth

Japan AI Consulting (J-AIC)

📅 Jul 2024 · Aug 2025 📍 Osaka · Yokohama

- Built **LLM-powered prototypes** for the consulting practice; re-deployed at JICA's TICAD9 booth in Yokohama.

PROJECTS

- **D'isum sensor pipelines:** IoT sensor stream architecture with ML-integration layer feeding precision analytics.
- **EKKOU platform:** full-stack AI platform (Flutter mobile, Flask / FastAPI backend, GPU-backed model serving).

Public code: github.com/mesabo

PUBLICATIONS & HONOURS

- **IEEE VTS Tokyo / Japan Chapter 2024 Student Paper Award** for the CNN-BiLSTM IoT load-forecasting paper (*VTC2024-Fall*, Washington D.C.).
- 2 first-author Q1 journals (*Internet of Things* IF 7.6, *ESWA* IF 7.5) and 5 IEEE flagship papers (VTC, GLOBE-COM, ICC, ICCE).
- JICA Development Studies Program graduate, 2025.

LANGUAGES

French ● ● ● ● ● English ● ● ● ● ● Japanese ● ● ● ● ●